

Unit 1: Fundamentals of Cloud Computing

(Weightage marks: 10marks)

Syllabus

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.
1	TLO 1.1 Explain characteristics of Cloud computing. TLO 1.2 Compare Cloud deployment models on the given services. TLO 1.3 Explain the given service offered by identified Cloud service model. TLO 1.4 Explain components of Cloud computing architecture.	Unit - I Fundamentals of Cloud Computing 1.1 Definition of Cloud Computing, Characteristics of Cloud computing 1.2 Cloud Deployment Models(Introduction, advantages and disadvantages) : Public Cloud, Private Cloud, Community Cloud, Hybrid Cloud 1.3 Cloud Service Models (Function, advantages, disadvantages) : IaaS, PaaS, SaaS 1.4 Cloud cost benefits 1.5 Architectural and Infrastructural components of Cloud Computing

Definition of Cloud Computing

- “Cloud Computing delivers computing services—servers, storage, databases, networking, software, and more—over the Internet (Cloud).”
- Cloud computing means using online servers to store, process, or access data instead of using your own computer or local server.
- Cloud Computing is the delivery of computing services like servers, storage, databases, networking, software, and more, over the Internet (Cloud).

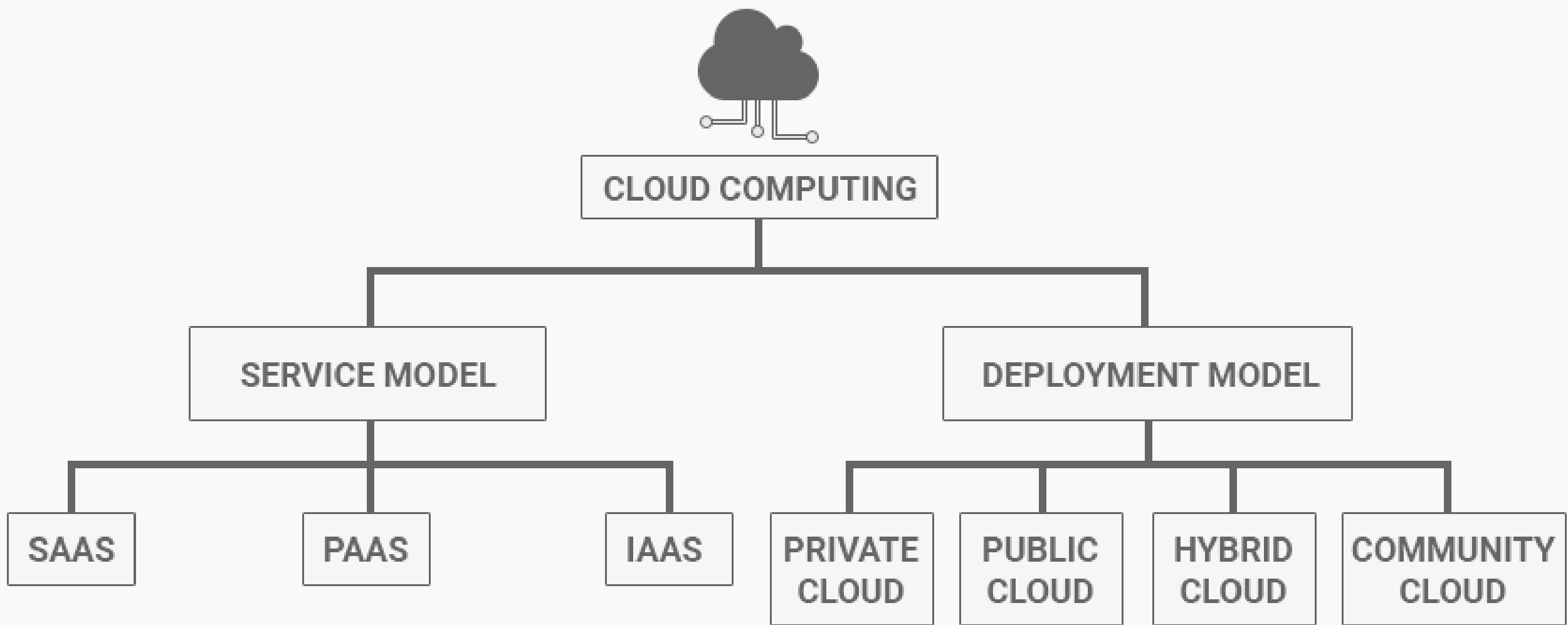


Characteristics of Cloud computing (2M)

Characteristic	Description
On-Demand Self-Service	Get resources anytime without human help
Broad Network Access	Accessible from anywhere via Internet
Resource Pooling	Shared resources among many users
Rapid Elasticity	Quick increase or decrease in resource usage
Measured Service	Pay only for what you use
Security	Data protection by the cloud provider
Automatic Maintenance	No manual updates or server repairs needed
Multi-Tenancy	One system used by many customers safely

Enlist any 4 cloud platform . (2M)

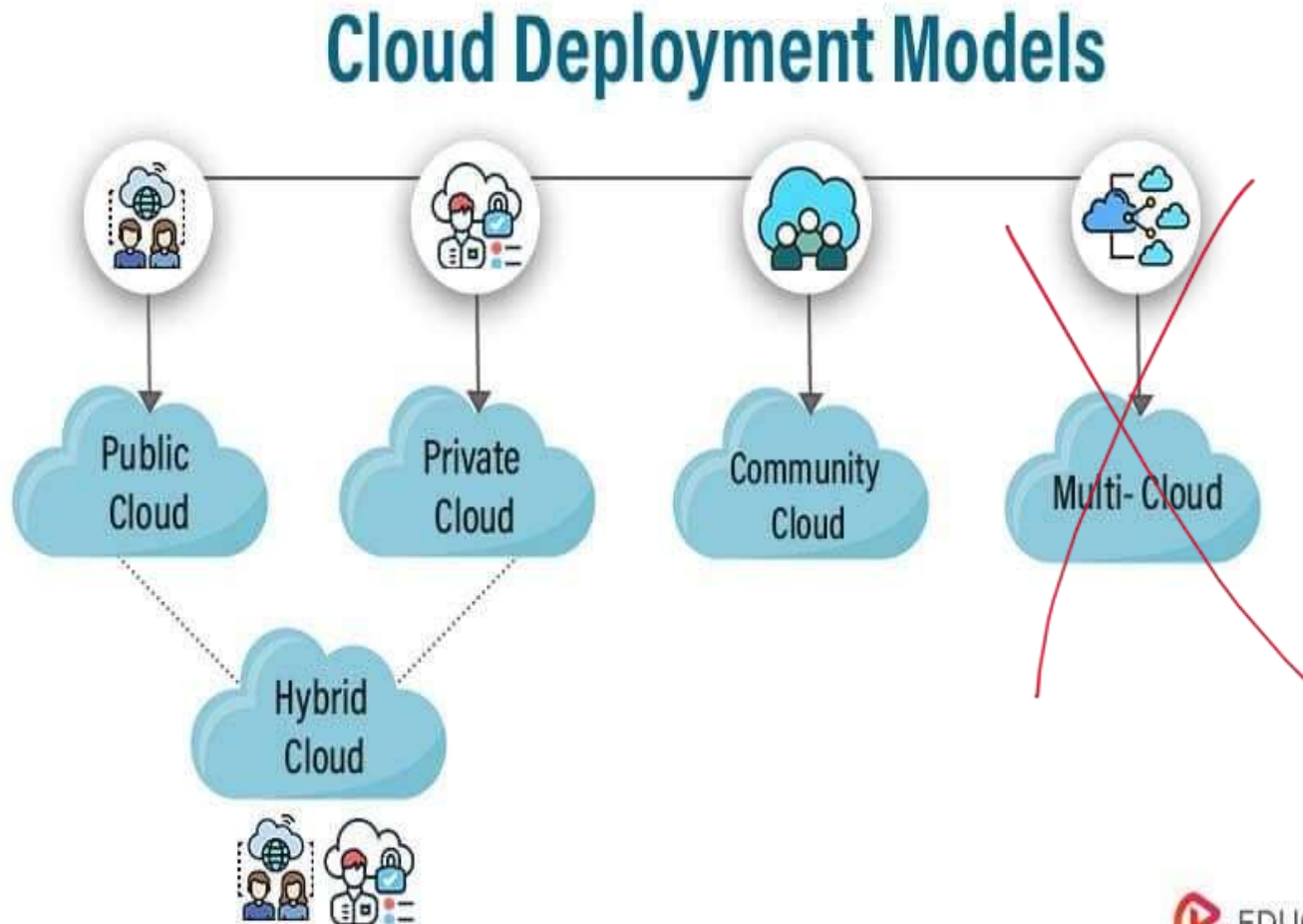
1. Amazon Web Services (AWS)
2. Microsoft Azure
3. Google Cloud Platform (GCP)
4. IBM Cloud
5. Oracle Cloud
6. Alibaba Cloud
7. Salesforce Cloud
8. DigitalOcean
9. Red Hat OpenShift
10. VMware Cloud



Questions asked :

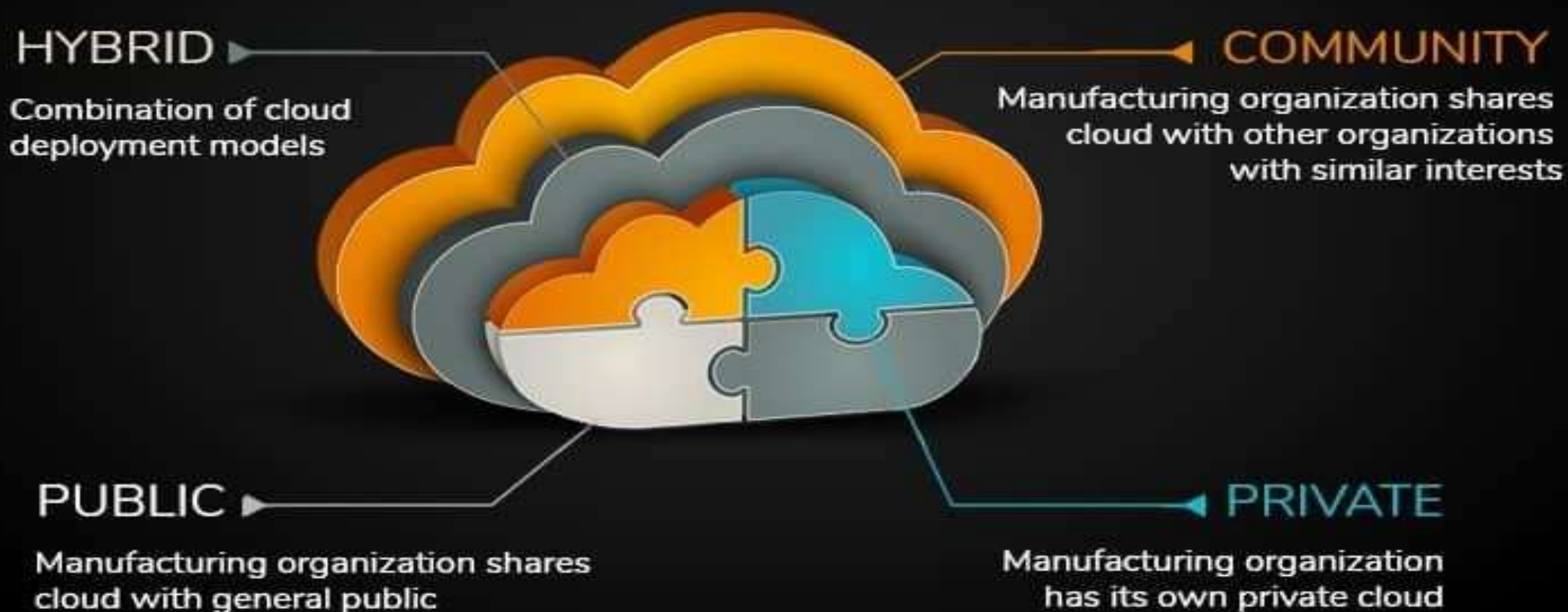
- 1) Differentiate between Public cloud and Private cloud (4m)
- 2) List any two types of cloud deployment model or Enlist cloud deployment models (2m)
- 3) Explain Paas.(2m)

☁ Cloud Deployment Models



- Where your cloud infrastructure is located.
- Who owns and manages it.
- How it will be used by users or organizations.
- It determines how **secure**, **scalable**, **flexible**, and **cost-effective** your cloud setup will be.

Cloud Deployment Models



Public Cloud Deployment Model

- The Public Cloud is a type of cloud computing in which services and infrastructure are provided over the Internet to the general public or large industry groups.
- The cloud infrastructure is owned and managed by a cloud service provider, not by the user.
- It allows anyone to access computing resources like servers, storage, and applications on a pay-per-use basis.
- 📌 Example: Google App Engine, Microsoft Azure, Amazon Web Services (AWS)

Enterprise P



Cloud Service Provider



Enterprise Q



User R

✓ Advantages:

- No setup cost or maintenance
- Pay-as-you-use
- Easily scalable
- Infrastructure managed by the provider

✗ Disadvantages:

- Less secure
- Limited customization
- Shared resources

Private Cloud Deployment Model

- The Private Cloud is the opposite of the public cloud. It provides a dedicated environment for a single organization or user. Resources (like servers, storage, etc.) are not shared with others.
- Also known as the "internal cloud", it is managed within the organization's boundary and offers high security and full control.
- The system is protected by strong firewalls and is usually maintained by the organization's IT team.
- 📌 Example: Company-owned data centers, VMware Private Cloud

On premise Private cloud

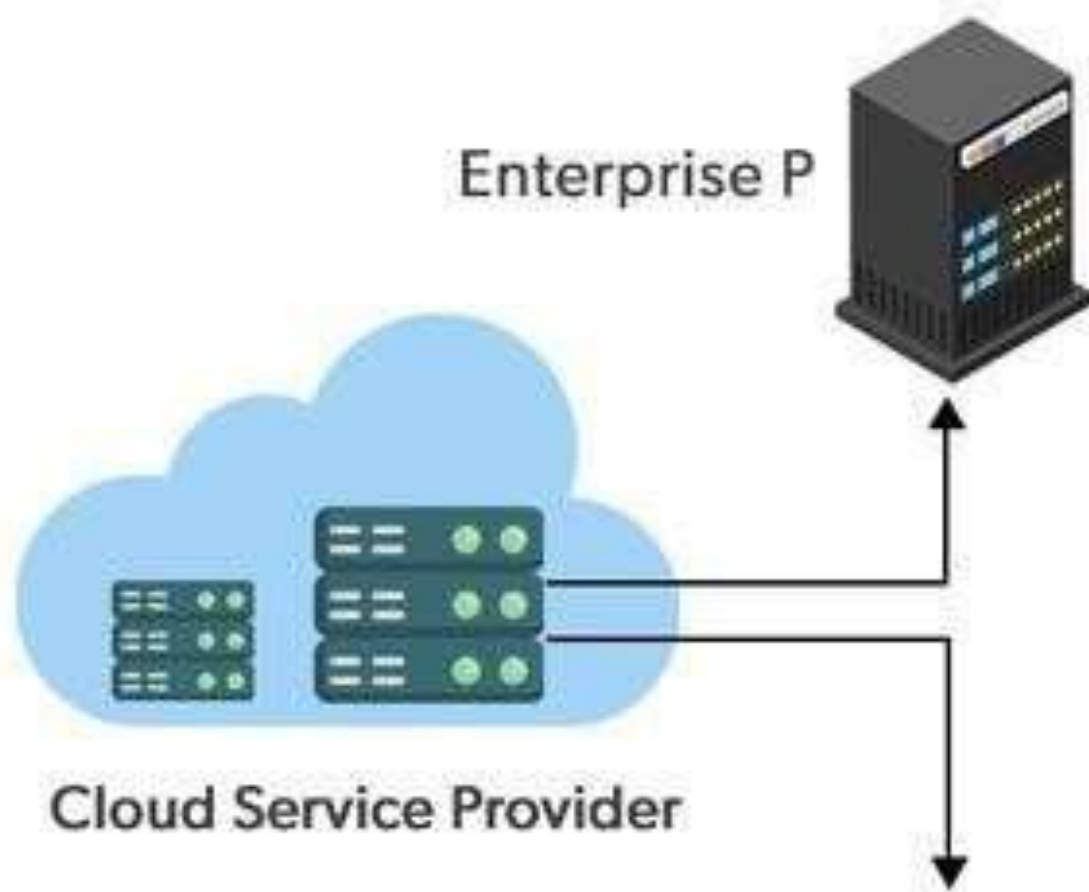
Enterprise P



Cloud Service Provider

Externally hosted Private cloud

Enterprise P



Cloud Service Provider

Dedicated for Enterprise P

Advantages & Disadvantages

✓ Advantages of the Private Cloud Model:

Better Control:

Full control over hardware, software, and security policies.

Data Security and Privacy:

High-level security with access only to authorized users.

Customization:

Can be tailored to meet specific needs of the organization.

Supports Legacy Systems:

Good for old systems that can't run on public cloud.

✗ Disadvantages of the Private Cloud Model:

Costly:

High initial investment in hardware and setup.

Less Scalable:

Scaling is limited to available physical infrastructure.

Requires IT Staff:

Needs in-house experts to maintain and manage systems.

Hybrid Cloud Deployment Model

- The Hybrid Cloud combines both **public and private cloud models**, allowing organizations to use the benefits of both.
- It uses **proprietary software** to connect these clouds so that **data and applications can move easily** between them.
- This model helps companies **host sensitive data** in the **private cloud** while using the **public cloud** for less critical operations — offering **cost savings** and **security together**.
- 📌 **Example:** Company stores customer records on a private server but uses AWS for hosting the company website.

Private cloud

Enterprise P



Cloud Service Provider



Public cloud

Enterprise P



Enterprise Q



Cloud Service Provider

Advantages and Disadvantages

Advantages of the Hybrid Cloud Model:

Flexibility and Control:

Businesses can design custom solutions by mixing public and private clouds.

Cost Efficiency:

Pay only for the extra resources used in the public cloud (scalable on demand).

Improved Security:

Sensitive data stays in a secure private cloud while still using public resources when needed.

✗ Disadvantages of the Hybrid Cloud Model:

Difficult to Manage:

Managing two different environments (public + private) is complex.

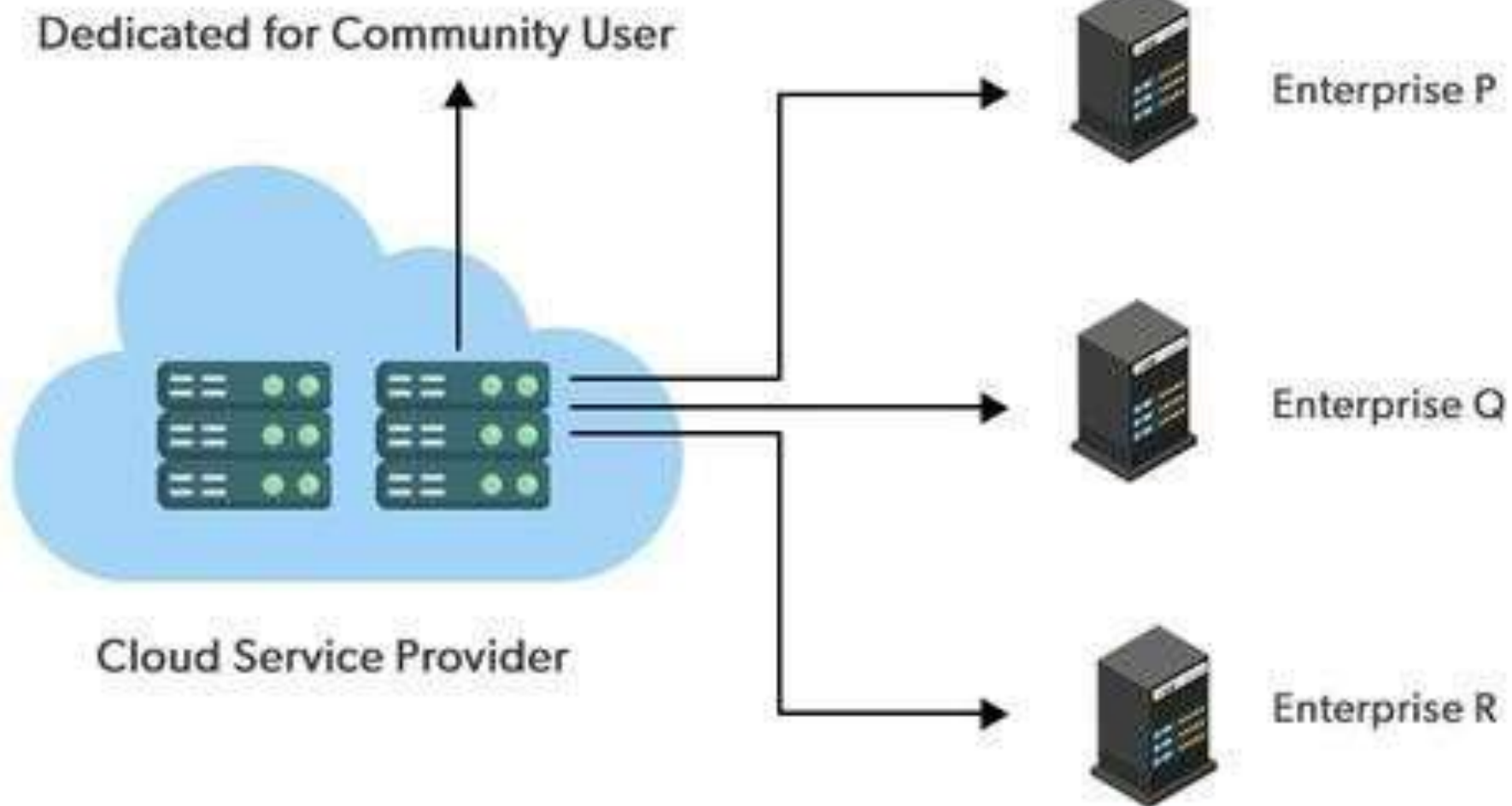
Slow Data Transmission:

Data moves through the public cloud, which can cause latency or delays.

Community Cloud Deployment Model

- The Community Cloud is a cloud model where the infrastructure is shared by multiple organizations that have similar goals, policies, or security requirements.
- It is a distributed system formed by combining services from different clouds to support a specific community, industry, or business group.
- The cloud can be managed by a third party or jointly managed by the participating organizations.
- 📌 **Example:** Educational institutions, government agencies, or banks using a shared cloud.

Community Users



Advantages and Disadvantages

✓ Advantages of the Community Cloud Model:

Cost Effective:

The cost is shared among several organizations, making it affordable.

Security:

Better security than public cloud due to restricted access.

Shared Resources:

Infrastructure and services can be shared efficiently among members.

Collaboration & Data Sharing:

Ideal for projects that require collaboration and secure data sharing among trusted parties.

✗ Disadvantages of the Community Cloud Model:

Limited Scalability:

Resource scaling is limited as multiple organizations are sharing the same infrastructure.

Rigid Customization:

Customizing services to one organization's specific needs is difficult, as it may affect others in the community.

(Questions) Home work !!!

1. Difference between Public and Private Cloud.
2. Explain Hybrid cloud with its applications.
3. What is community cloud ?
4. Advantages or disadvantages of any cloud (public ,private , hybrid or community)

Cloud Service Models

- These are sometimes called the **cloud computing stack** because they are built on top of one another. Knowing what they are and how they are different, makes it easier to accomplish your goals. These abstraction layers can also be viewed as a layered architecture.





SaaS
Software as a Service

EIM
Incidents
EHS
Tasks
Waste

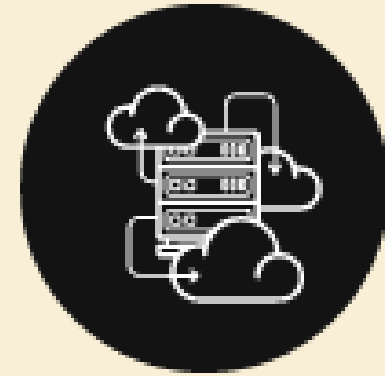
Use it



PaaS
Platform as a Service

App Development
Messaging
Dashboards
Integration

Build with it



IaaS
Infrastructure as a Service

Networking
Security
System Management
Scalability

Move to it

1. Software as a Service(SaaS)

- Software-as-a-Service (SaaS) means using software over the internet instead of installing it on your computer. You don't have to worry about downloading, updating, or maintaining anything- the company that provides the software handles all of that.

Example:

- Think of **Google Docs**. You don't need to install it. You just open your browser, log in, and start using it. Google stores your work and keeps the software updated. You just use it when you need it.
- SaaS is usually offered on pay-as-you-go basis, and you can access it from any device with internet. It's also called **web-based-software** or **on-demand software** because you can use it anytime, anywhere, without setup

Examples of SaaS Providers

- Salesforce.com
- Microsoft Office 365
- Dropbox
- Cloud9 Analytics
- BigCommerce
- Eloqua
- CloudSwitch
- CloudTran

Advantages and Disadvantages

Advantages of SaaS

1. **Saves cost** - You only pay for what you use without buying full software.
2. **No installation needed** - Apps run directly from the browser without setup.
3. **Access from anywhere** - Use the software from any location with internet.
4. **Updates happen automatically** - The provider handles all software updates.
5. **Easily scalable** - You can quickly upgrade services as your needs grow.

✗ Disadvantages of SaaS

1. **Limited customization** - You can't fully modify the software features.
2. **Needs internet always** - Software won't work well without a good connection.
3. **Risk of data breaches** - Data is stored online and can be attacked.
4. **Less control over data** - The provider may access or manage your data.

2. Platform as a Service

- [PaaS](#) is a type of cloud service that gives developers the tools they need to build and launch apps online without setting up any hardware and software themselves.
- With PaaS, everything runs on the provider's server and is accessed through a web browser. The provider takes care of things like servers, storage, and operating systems. Developers just focus on writing and managing the app.

- **Example:**

Imagine you're planning a school's annual day event. You have two options.

1 .Build the Venue yourself (buy land, set up a stage, arrange lighting, etc.).

2.Or **rent a ready-to-use** venue and just focus on the actual event.

- PaaS is like renting the venue- it saves time, efforts, and setup costs, so you can completely focus on what matters: **building your app**.
- You don't control the back-end (like servers), but you do control the app you create and how it behaves.

Advantages and Disadvantages

✓ Advantages of PaaS

1. **Simple and convenient** - Users can access tools and infrastructure easily via a web browser.
2. **Cost-effective** - Pay only for what you use, avoiding hardware/software costs.
3. **Supports full app lifecycle** - Helps in building, testing, deploying, and updating apps.
4. **Improves efficiency** - Enables faster development with less complexity.

✗ Disadvantages of PaaS

1. **Limited infrastructure control** - Users can't fully control or customize the backend.
2. **Provider dependency** - Service relies completely on the provider's reliability.
3. **Limited flexibility** - May not support all workloads or special app needs

Examples of PaaS Providers

AWS Elastic Beanstalk, Salesforce, Microsoft Azure, Google App Engine, CloudBees, IBM SmartCloud.

3. Infrastructure as a Service

- Infrastructure as a service (IaaS) is a cloud service where companies rent IT resources like servers, storage, and networks instead of buying and managing them.
- It's like outsourcing your computer hardware. The cloud provider gives you the basic building blocks (like virtual machines, storage, and internet access), and you use them to run your apps and services.
- You pay based on how much you use - by the hour, week, or month. The way, you don't need to spend a lot of money on buying hardware.
- **Example:**
- Imagine you want to start a website. Instead of buying you own server, you rent on a cloud provider's server. You use their storage and networking, but you control what runs on it-like your website or app.
- That's IaaS, You get the flexibility and power of your own setup, without the cost and trouble of maintaining hardware.

Advantages and Disadvantages

✓ Advantages of IaaS

Cost-effective - No need to buy hardware; pay only for what you use (hourly, weekly, or monthly).

Website hosting - Hosting websites is cheaper than traditional web hosting methods.

Better security - Cloud providers may offer stronger security than local systems.

No maintenance - Provider handles infrastructure updates and data center management.

✗ Disadvantages of IaaS

Limited control - Users can't fully customize or manage the backend infrastructure.

Security responsibility - Users must secure their own apps and data.

Access issues - May not be available in certain countries due to legal restrictions.

• Examples of IaaS Providers

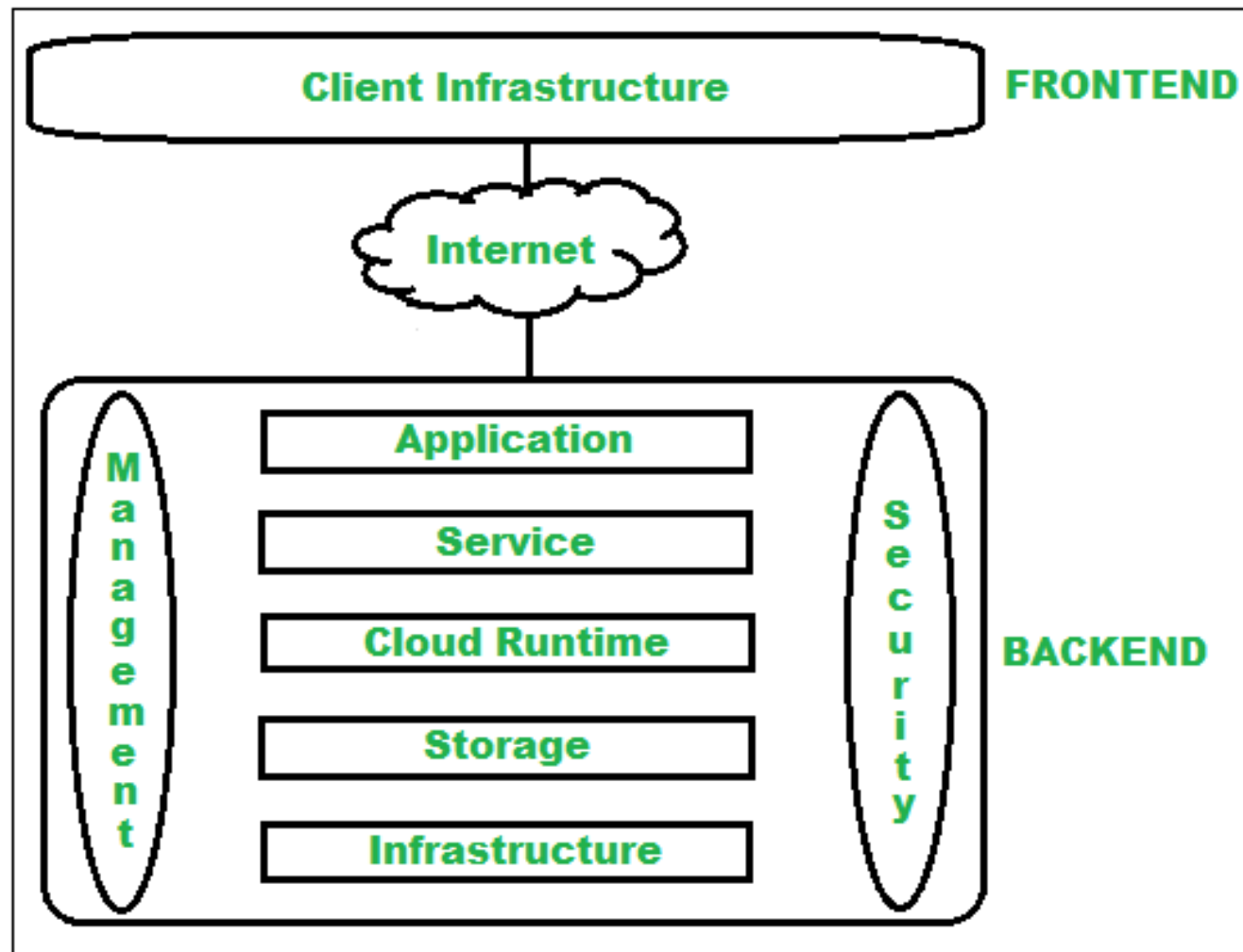
- Amazon Web Services, Bluestack, IBM, OpenStack, Rackspace, VMware.

Cloud cost benefits (2MARKS)

- **Pay-as-you-go** - Only pay for what you use (no upfront investment).
- **No hardware cost** - No need to buy or maintain physical servers.
- **Reduced maintenance** - Providers handle updates, saving time and labor costs.
- **Scalability** - Easily scale up/down without buying new infrastructure.
- **Energy savings** - Lower power and cooling costs compared to on-premises data centers.
- **Less IT staff needed** - Fewer people required to manage infrastructure.
- **Global reach at low cost** - Access services worldwide without setting up offices.
- **No software licensing hassle** - Software is bundled in the cloud service cost.

Architecture of Cloud Computing (4MARKS FIX)

- Architecture of cloud computing is the combination of both SOA (Service Oriented Architecture) and EDA (Event Driven Architecture). Client infrastructure, application, service, runtime cloud, storage, infrastructure, management and security all these are the components of cloud computing architecture.
- The cloud architecture is divided into 2 parts, i.e.
 - Frontend
 - Backend



1. Frontend

- The client-side of cloud computing.
- Includes user interfaces and applications used to access cloud services.
- Example: Using a web browser to access Google Drive.

2. Backend

The cloud itself, managed by the provider.

Includes all resources, services, and security mechanisms.

Manages storage, VMs, applications, deployment, and more.

Components of Cloud Computing Architecture

Frontend Component

- **Client Infrastructure:**
Provides a GUI (Graphical User Interface) for users to interact with the cloud.

Backend Components

- **Application**
 - Software/platform that provides services to the user.
- **Service**
 - Manages the three main service models: SaaS, PaaS, IaaS.
- **Runtime Cloud**
 - Provides execution environment/platform for running virtual machines.
- **Storage**
 - Offers scalable storage services for data saving and management.

- **Infrastructure**
Includes **servers, storage, network devices, virtualization tools.**
- **Management**
Controls and manages all backend components and resources.
- **Security**
Implements security mechanisms to protect cloud systems and user data.
- **Database**
 - Stores structured/unstructured data (e.g., **SQL, NoSQL**).
 - Examples: Amazon RDS, Google Cloud SQL.
- **Networking**
Supports **load balancing, DNS, VPNs, and data routing.**
- **Analytics**
Provides tools for **data analysis, business intelligence, machine learning.**
- **Internet**
Acts as a **bridge** between frontend and backend for communication.

Infrastructural Components of Cloud Computing

Component	Description
Hardware	Physical servers, data centers, and storage systems used to run cloud services.
Virtualization	Enables multiple virtual machines to run on a single physical system.
Storage	Centralized cloud-based storage systems like SAN (Storage Area Network) or NAS (Network Attached Storage).
Network	High-speed internet connectivity and routing for data transfer.
Service Models	Cloud service types: SaaS , PaaS , and IaaS .
Deployment Models	Cloud types: Public , Private , Hybrid , and Community Cloud .



THANK YOU!